

Applied NLP

DATA 641

Lecture 4 — Text Representation; Bag of words

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Bag of words

- Bag of words (BoW) is a classical text representation technique that has been used commonly in NLP, especially in text classification problems.
- The key idea behind it is as follows: represent the text under consideration as a bag (collection) of words while ignoring the order and context. The basic intuition behind it is that it assumes that the text belonging to a given class in the dataset is characterized by a unique set of words.
- Similar to one-hot encoding, BoW maps words to unique integer IDs between 1 and $|V|$. Each document in the corpus is then converted into a vector of $|V|$ dimensions where in the i th component of the vector, $i = w_{id}$, is simply the number of times the word w occurs in the document.
- For our toy corpus, where the word IDs are dog = 1, bites = 2, man = 3, meat = 4, food = 5, eats = 6, D1 becomes $[1 \ 1 \ 1 \ 0 \ 0 \ 0]$. This is because the first three words in the vocabulary appeared exactly once in D1, and the last three did not appear at all. D4 becomes $[0 \ 0 \ 1 \ 0 \ 1 \ 1]$.

D1	Dog bites man.
D2	Man bites dog.
D3	Dog eats meat.
D4	Man eats food.

Some of the **advantages** of this encoding include:

- Like one-hot encoding, BoW is fairly simple to understand and implement.
- With this representation, documents having the same words will have their vector representations closer to each other in Euclidean space as compared to documents with completely different words. The distance between D1 and D2 is 0 as compared to the distance between D1 and D4, which is 2. Thus, the vector space resulting from the BoW scheme captures the semantic similarity of documents. So if two documents have similar vocabulary, they will be closer to each other in the vector space and vice versa.
- We have a fixed-length encoding for any sentence of arbitrary length.

Some of the **disadvantages** of this encoding include:

- The size of the vector increases with the size of the vocabulary. Thus, sparsity continues to be a problem. One way to control it is by limiting the vocabulary to n number of the most frequent words.
- It does not capture the similarity between different words that mean the same thing. Say we have three documents: "I run", "I ran", and "I ate". BoW vectors of all three documents will be equally apart.
- This representation does not have any way to handle out of vocabulary words (i.e., new words that were not seen in the corpus that was used to build the vectorizer).
- As the name indicates, it is a "bag" of words—word order information is lost in this representation. Both D1 and D2 will have the same representation in this scheme.